	GEETHANJALI INSTITUTE OF SCIENCE & TECHNOLOGY **(AN AUTONOMOUS INSTITUTION)** **(Approved by AICTE, New Delhi & Affiliated to JNTUA, Ananthapuramu)** **(Accredited by NAAC with "A" Grade, NBA (EEE,ECE &ME) & ISO9001:2008CertifiedInstitution)**

QUESTIONBANK(DESCRIPTIVE)

Subject Name with Code: Cryptography &Network Security(23A0526T)

Course & Branch: B.TECHCSE(CS), AIML,CSE Year& Semester: III-IIRegulation: RG23

UNIT - I

S.No.	Question	[BT Level] [CO][Marks]
2 Marks Questions (Short)		
1.	State the **block size and key sizes of AES**.	L1/CO1/2M
2.	What is a **Transposition Cipher**?	L1/CO1/2M
3.	Define the **Symmetric Cipher Model**.	L1/CO1/2M
4.	What is meant by a **Security Attack**?	L1/CO1/2M
5.	What are **Security Mechanisms**? Give two examples.	L1/CO1/2M
6.	List out the Various types of **Security Services**.	L1/CO1/2M
7.	Distinguish between **Passive and Active attacks**	L2/CO1/2M
8.	What is **Steganography**?	L2/CO1/2M
9.	What is **OSI Security Architecture**?	L1/CO1/2M
10.	Define **Computer Security** and state its objectives.	L2/CO1/2M
11.	**Explain OSI Security Architecture**. Describe the components of the architecture and their significance.	L2/CO1/10M
12.	Write short notes on i) cryptography and ii) Steganography	L2/CO1/10M
13.	**Explain Security Attacks**. Classify them into **passive and active attacks** with suitable examples.	L2/CO1/10M
14.	**Explain Security Services** provided by network security. Describe authentication, confidentiality, integrity, and non-repudiation	L2/CO1/10M
15.	**Explain Security Mechanisms**. Discuss how these mechanisms support security services.	L2/CO1/10M
16.	**Explain the Model for Network Security** with a neat diagram and describe the role of encryption and keys.	L4/CO1/10M
17.	**Explain Classical Encryption Techniques**. Describe **substitution techniques and transposition techniques** with examples.	L2/CO1/10M
18.	Explain the encrypt the message "PAY" using hill cipher with the following key matrix and show the decryption to get original plain text. $k = \begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$	L2/CO1/10M
19.	**Explain the Traditional Block Cipher Structure** and describe the **Data Encryption Standard (DES)** in detail.	L2/CO1/10M
20.	**Explain the Advanced Encryption Standard (AES)**. Describe the **AES structure and transformation functions**.	L2/CO1/10M

UNIT - II

S.No	Question	[BT Level] [CO][Marks]
1	Define the Euclidean Algorithm.	L1/CO2/2M
2	What is meant by Modular Arithmetic?	L1/CO2/2M
3	State Fermat's Little Theorem.	L3/CO2/2M
4	State Euler's Theorem.	L1/CO2/2M
5	What is the Chinese Remainder Theorem?	L1/CO2/2M
6	Define Discrete Logarithm.	L1/CO2/2M
7	What is a Finite Field?	L1/CO2/2M
8	Define a finite field of the form GF(p).	L2/CO3/2M
9	What is Public Key Cryptography?	L1/CO3/2M
10	What is Elliptic Curve Cryptography (ECC)?	L1/CO3/2M

S.No	Question	[BT Level] [CO][Marks]
11	Explain Fermat's Little Theorem and Euler's Theorem, and describe their significance in modular computations.	L1/CO2/10M
12	State Chinese Remainder Theorem and find X for the given set of congruent equations using CRT: (a) $X \equiv 1 \pmod 5$ $X \equiv 2 \pmod 7$ $X \equiv 3 \pmod 9$ b) $X \equiv 2 \pmod 3$ $X \equiv 3 \pmod 5$ $X \equiv 2 \pmod 7$ $X \equiv 4 \pmod{11}$	L2/CO2/10M
13	Explain ECC – Diffie-Hellman Key Exchange with both keys in detail with an example.	L4/CO2/10M
14	Analyze the Discrete Logarithm Problem and explain why it is computationally hard.	L2/CO2/10M
15	Explain the RSA algorithm to perform key generation, encryption, and decryption for secure communication and apply RSA algorithm to the following example: p = 7, q = 11, e = 17, M = 8.	L3/CO3/10M
16	Explain the concept of Finite Fields and describe finite fields of the form GF(p) and GF(2^n).	L3/CO2/10M
17	Analyze the security of RSA based on the integer factorization problem. Explain it.	L4/CO3/10M
18	Analyze the role of modular arithmetic and finite fields in public key cryptographic algorithms. Briefly explain it.	L3/CO2/10M
19	User Alice and Bob exchange the key using Diffie-Hellman algorithm. Assume $\alpha = 5$, q = 83, XA = 6, XB = 10. Find YA, YB, and K.	L2/CO2/10M

UNIT - III

S.No	Question	[BT Level] [CO][Marks]
1	What is a Cryptographic Hash Function?	L1/CO2/2M
2	List the basic requirements of a secure hash function.	L1/CO2/2M
3	What is the Secure Hash Algorithm (SHA)?	L1/CO2/2M
4	Define Message Authentication.	L1/CO2/2M
5	What is a Message Authentication Code (MAC)?	L1/CO2/2M
6	What is HMAC?	L1/CO2/2M
7	What is CMAC?	L1/CO2/2M
8	Define Digital Signature.	L1/CO2/2M
9	What is an X.509 Certificate?	L1/CO2/2M
10	What is Public Key Infrastructure (PKI)?	L1/CO2/2M

11.	Explain Cryptographic Hash Functions with Applications, Requirements & Security in Cryptography?	L1/CO3/10M
12.	Explain the Secure Hash Algorithm (SHA) Family with a neat Sketch?	L2/CO3/10M
13.	Explain Message Authentication Functions with Requirements & Security Application?	L4/CO3/10M
14.	Explain Hash-based Message Authentication Codewith Complete Working Process and Security Analysis?	L2/CO3/10M
15.	Explain Cipher-based Message Authentication Codewith Complete Working and Advantages?	L3/CO3/10M
16.	Explain Digital Signatures and Their Security Services with a nreat Sketch?	L3/CO3/10M
17.	Explain NIST Digital Signature Algorithm (DSA) with Complete Working Process?	L4/CO3/10M
18.	Discuss Distribution of Public Keys &its Importance with a neat Sketch?	L3/CO3/10M
19.	Explain X.509 Certificates &its Structure and Working Process Implementation?	L2/CO3/10M
20.	Explain Public Key Infrastructure (PKI) Framework with a neat Sketch?	L2/CO3/10M
21.	Differentaiate HMAC&CMAC Security Analysis in Real Time?	L2/CO3/10M

Unit-IV

S.No	Question	[BT Level] [CO][Marks]
1	What is Remote User Authentication?	L1/CO4/2M
2	What is meant by Authentication Protocol?	L1/CO4/2M
3	What is SPD?	L1/CO4/2M
4	What are the main components of Kerberos?	L1/CO4/2M
5	What is a Ticket in Kerberos authentication?	L1/CO4/2M
6	What is Pretty Good Privacy (PGP)?	L1/CO4/2M
7	What is S/MIME?	L1/CO4/2M
8	What is IP Security (IPSec)?	L1/CO4/2M
9	What is Encapsulating Security Payload (ESP)?	L1/CO4/2M
10	What is Internet Key Exchange (IKE)?	L1/CO4/2M

11.	Explain Remote User Authentication Principles with a neat Sketch?	L1/CO4/10M
12.	Explain Kerberos Authentication Protocol&its Working Process?	L2/CO4/10M
13.	Explain the Working Process of Pretty Good Privacy (PGP) ?	L4/CO4/10M
14.	Explain S/MIME (Secure/Multipurpose Internet Mail Extensions)&Its Working Process?	L2/CO4/10M
15.	Explain IP Security Policy and Security Associations with a neat Sketch?	L3/CO4/10M
16.	Explain Encapsulating Security Payload (ESP)with a neat Sketch&its Working Process?	L3/C04/10M
17.	Explain Combining Security Associations in IPSecurity &its Types in realtime?	L4/CO4/10M
18.	Explain Internet Key Exchange (IKE)&its Components and Phases in realtime Implementation?	L3/CO4/10M
19.	Compare PGP, S/MIME and IPSec with a neat Sketch?	L2/CO4/10M
20.	Write a short note on IP Security (IPSec) Overview &its Applications?	L2/CO4/10M

Unit-V

S.No	Question	[BT Level] [CO][Marks]
1	What is Remote User Authentication?	L1/CO5/2M
2	What is meant by Authentication Protocol?	L1/CO5/2M
3	What is Kerberos?	L1/CO5/2M
4	What are the main components of Kerberos?	L1/CO5/2M
5	What is a Ticket in Kerberos authentication?	L1/CO5/2M
6	What is Pretty Good Privacy (PGP)?	L1/CO6/2M
7	What is S/MIME?	L1/CO6/2M
8	What is IP Security (IPSec)?	L1/CO6/2M
9	What is Encapsulating Security Payload (ESP)?	L1/CO6/2M
10	What is Internet Key Exchange (IKE)?	L1/CO6/2M

11.	Explain Web Security Requirements in detail?with a neat Sketch?	L1/CO5/10M
12.	Explain Transport Layer Security (TLS) Protocol &itys Sub Protocol sin Detail?	L2/CO5/10M
13.	Explain TLS Record Protocol Format and its Working Process?	L4/CO5/10M
14.	Explain HTTPS Protocol Stack &its Security Services in Realtime?	L2/CO5/10M
15.	Explain Secure Shell (SSH) Protocol&its Working Process in detail?	L3/CO5/10M
16.	Explain Firewall Characteristics and its Access Policy elements in detail?	L3/CO6/10M
17.	Explain Various Types of Firewalls &its Working Process?	L4/CO6/10M
18.	Explain Firewall Configurations &its Types in real Time?	L3/CO6/10M
19.	Compare TLS, HTTPS and SSH Protocols with a neat Sketch?	L2/CO6/10M
20.	Discuss Firewall Operation and its Importance in Realtime Implementation?	L2/CO6/10M

Signature of the Staff:

Signature of Department Academic Committee Member 1:

Signature of Department Academic Committee Member 2:

Signature of Department Academic Committee Member 3: